Number Sequences (Set B)

Year 4 Pt 2 · **Mark Scheme**

Lesson Worksheet — Answers

Spotting the Pattern

1. 15 — gap +3 throughout (3, 6, 9, 12, 15).
2. 30 — gap -5 throughout (50, 45, 40, 35, 30).
3. D (32) — each term doubles (2, 4, 8, 16, 32).
4. A, C, E — these all increase by exactly 4 each time; B increases by 4,4,5 and D increases by 4,5,4, so they are inconsistent.
5. 35 — gap +7 throughout (7, 14, 21, 28, 35).
6. C (60) — gap -10 throughout (100, 90, 80, 70, 60).
7. 36 — square numbers: 1^2, 2^2, 3^2, 4^2, 5^2, 6^2 = 1, 4, 9, 16, 25, 36.
8. 15 — 5, 10, 15, 20, 25 (gap +5 throughout).
9. D (32) — each term doubles (1, 2, 4, 8, 16, 32).
10. A, C, D — each term is multiplied by 3 (geometric), so the next term is 81×3=243; it is not arithmetic, and the increase is not a constant 6 (it increases by 6, 18, 54).
11. 14 — gap +3 throughout (2, 5, 8, 11, 14).
12. C (13) — Fibonacci rule: 5+8=13.
13. 36 — gap -6 throughout (60, 54, 48, 42, 36).
14. 16 — each term doubles: 2, 4, 8, 16, 32.
15. A, B, D — these are the first five square numbers, the next term is 36 (6^2), and the gap between terms increases each time (3, 5, 7, 9); the sequence does not increase by a constant amount, and it is increasing, not decreasing, so C and E are false.

Applying the Pattern

16. 39 — gaps increase by 2 each time: 4→7(+3), 7→12(+5), 12→19(+7), 19→28(+9), 28→39(+11).
17. 6.25 — each term is halved: 100, 50, 25, 12.5, 6.25.
18. 21 — Fibonacci-style: 8+13=21.
19. 17 — 3→5 (3×2−1=5), 5→9 (5×2−1=9), 9→17 (9×2−1=17).
20. A, C, E — each term is the sum of the two preceding terms; B and D increase by a constant amount instead (arithmetic, not Fibonacci-style).
21. 50 — gaps increase by 3 each time: 5→8(+3), 8→14(+6), 14→23(+9), 23→35(+12), 35→50(+15).
22. 1, 5, 13, 29 — term1=1; term2=1×2+3=5; term3=5×2+3=13; term4=13×2+3=29.
23. 21 — triangular numbers: each term adds the next counting number (1, +2=3, +3=6, +4=10, +5=15, +6=21).
24. The gap increases by 2 each time (+3, +5, +7, +9); the next term is 27+9=36.
25. A, C, E — in each of these, the gaps themselves increase by a constant amount; B has a constant gap (not increasing), and D's gaps double rather than increase by a fixed amount.
26. 12.5 — each term is halved: 200, 100, 50, 25, 12.5.
27. 7, 12, 19 — 2+5=7, 5+7=12, 7+12=19.
28. 720 — each term is multiplied by an increasing amount: 1×2=2, 2×3=6, 6×4=24, 24×5=120, 120×6=720.
29. A, C, D — each has a constant multiplying ratio (×2, ×0.5, ×3 respectively); B increases by a constant amount (arithmetic, not geometric) and E also increases by a constant amount (+2 each time).
30. 16, 28 — 10−2=8, 8×2=16; 16−2=14, 14×2=28.

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Sequences in Context

31. 95 — the rule is "double the previous term, then add 1": 2×2+1=5, 5×2+1=11, 11×2+1=23, 23×2+1=47, 47×2+1=95.

32. 4, 5, 7, 11, 19 — 4×2−3=5, 5×2−3=7, 7×2−3=11, 11×2−3=19.

33. A, B, D — the gaps increase by 1 each time (1,2,3,4,5), so the next gap is 6 and the next term is 16+6=22, making B true; D is true. The sequence is not geometric, so C is false; the 7th term is 22, not 23, so E is false.

34. 1, 2, 3, 6, 5, 10, 7, 14 — alternating the odd sequence (1,3,5,7) with the even sequence (2,6,10,14), taken one at a time.

35. 47 — this is a Fibonacci-style rule where each term is the sum of the two before it: 3+4=7, 4+7=11, 7+11=18, 11+18=29, 18+29=47.

36. 100, 51, 26.5, 14.25, 8.125 — each term is halved then increased by 1: 100÷2+1=51, 51÷2+1=26.5, 26.5÷2+1=14.25, 14.25÷2+1=8.125.

37. A, B, D — these are cube numbers, so the correct next term is 125 (5^3), not 100; the gaps (7, 19, 37) do not form a constant arithmetic increase, ruling out a simple pattern-based guess of 100. The sequence is not arithmetic, so E is false, and C's reasoning is incorrect.

38. 5, 7, 10, 14, 19 — term2=5+2=7, term3=7+3=10, term4=10+4=14, term5=14+5=19.

39. The gaps are 4, 6, 8, 10 — increasing by 2 each time, not constant, so it is not arithmetic; the ratios (6÷2=3, 12÷6=2, 20÷12≈1.67) are not constant, so it is not geometric either. The actual rule: each term is 2×n×(n+1) for position n, or equivalently the gaps increase by 2 each time; the next term is 30+12=42.

40. 17 — alternating +6 and −2 throughout: 3+6=9, 9−2=7, 7+6=13, 13−2=11, 11+6=17.

41. These totals are square numbers (the sum of the first n odd numbers always equals n^2): 0,1,4,9,16,25… The 6th total is 25.

42. A, B, D, E — each term is half the previous (A), the next term is 5÷2=2.5 (B), this is geometric with ratio 1/2 (D, E); it is not arithmetic, so C is false.

43. 4, 6, 9 — term(n)=term(n−1)+term(n−2)−1: term3=3+2−1=4, term4=4+3−1=6, term5=6+4−1=9.

44. 33 — the gaps are 1, 2, 3, 5, 8 (each gap is the sum of the previous two gaps, Fibonacci-style); the next gap is 5+8=13, so the next term is 20+13=33.

45. This is the triangular number pattern (n people need n(n−1)/2 handshakes); for 6 people: 6×5/2=15.

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Homework Worksheet — Answers

Quick Practice

1. 25 — gap +5 throughout (5, 10, 15, 20, 25).

2. 50 — gap -10 throughout (90, 80, 70, 60, 50).

3. D (48) — each term doubles (3, 6, 12, 24, 48).

4. A, C, E — these increase by exactly 3 each time; B increases by 3,4,3 and D increases by 3,2,3, so they are inconsistent.

5. 40 — gap +8 throughout (8, 16, 24, 32, 40).

6. B (45) — gap -9 throughout (81, 72, 63, 54, 45).

7. 125 — cube numbers: 1^3, 2^3, 3^3, 4^3, 5^3 = 1, 8, 27, 64, 125.

8. 12 — 4, 8, 12, 16, 20 (gap +4 throughout).

9. D (96) — each term doubles (3, 6, 12, 24, 48, 96).

10. A, C, D — each term is multiplied by 3 (geometric), so the next term is 54×3=162; it is not arithmetic, and the increase is not constant (it increases by 4, 12, 36).

11. 24 — gap +5 throughout (4, 9, 14, 19, 24).

12. D (26) — Fibonacci rule: 10+16=26.

13. 42 — gap -7 throughout (70, 63, 56, 49, 42).

14. 24 — each term doubles: 3, 6, 12, 24, 48.

15. A, B, C — these are even numbers, the next term is 12, and each term increases by a constant amount of 2; the gap does not increase (it stays at 2), and the sequence is increasing, not decreasing, so D and E are false.

Mixed Practice

16. 33 — gaps increase by 2 each time: 3→5(+2), 5→9(+4), 9→15(+6), 15→23(+8), 23→33(+10).

17. 2 — each term is divided by 3: 162, 54, 18, 6, 2.

18. 23 — Fibonacci-style: 9+14=23.

19. 30 — 2→6 (2×2+2=6), 6→14 (6×2+2=14), 14→30 (14×2+2=30).

20. A, C, D — each term is the sum of the two preceding terms; B and E increase/multiply by a constant amount instead (not Fibonacci-style).

21. 51 — gaps increase by 3 each time: 6→9(+3), 9→15(+6), 15→24(+9), 24→36(+12), 36→51(+15).

22. 2, 4, 10, 28 — term1=2; term2=2×3−2=4; term3=4×3−2=10; term4=10×3−2=28.

23. 28 — triangular numbers: 1,3,6,10,15,21,28 (adding 1,2,3,4,5,6,7 in turn).

24. The gap increases by 2 each time (+3, +5, +7, +9); the next term is 26+9=35.

25. A, C, E — in each, the gaps themselves increase by a constant amount; B has a constant gap (not increasing) and D's gaps multiply by 3 rather than increasing by a fixed amount.

26. 5 — each term is divided by 3: 405, 135, 45, 15, 5.

27. 7, 11, 18 — 3+4=7, 4+7=11, 7+11=18.

28. 360 — each term is multiplied by an increasing amount: 1×3=3, 3×4=12, 12×5=60, 60×6=360.

29. A, C, D — each has a constant multiplying ratio (×2, ×0.5, ×3 respectively); B and E increase by a constant amount (arithmetic, not geometric).

30. 8, 22 — starting at 1: add 3 → 4, multiply by 2 → 8 (first new term); then add 3 → 11, multiply by 2 → 22 (second new term).

Standard / Advanced

31. 94 — the rule is "double the previous term, then add 2": 1×2+2=4, 4×2+2=10, 10×2+2=22, 22×2+2=46, 46×2+2=94.

32. 3, 7, 15, 31, 63 — 3×2+1=7, 7×2+1=15, 15×2+1=31, 31×2+1=63.

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33. A, B, D — the gaps increase by 1 each time (1,2,3,4,5), so the next gap is 6 and the next (7th) term is 17+6=23, confirming B; the sequence is not geometric, so C is false; the 7th term is 23, not 24, so E is false.

34. 2, 1, 4, 5, 6, 9, 8, 13 — alternating the first sequence (2,4,6,8) with the second sequence (1,5,9,13), taken one at a time.

35. 46 — the rule is "sum of the two previous terms, plus 1": 2+3+1=6, 3+6+1=10, 6+10+1=17, 10+17+1=28, 17+28+1=46.

36. 200, 102, 53, 28.5, 16.25 — each term is halved then increased by 2: 200÷2+2=102, 102÷2+2=53, 53÷2+2=28.5, 28.5÷2+2=16.25.

37. A, B, D — these are 4th powers, so the correct next term is 625 (5^4), not 400; the gaps (15, 65, 175) do not increase by a constant amount, ruling out a simple pattern-based guess of 400. The sequence is not arithmetic, so E is false, and C's reasoning is incorrect.

38. 3, 5, 8, 12, 17 — term2=3+2=5, term3=5+3=8, term4=8+4=12, term5=12+5=17.

39. The gaps are 6, 9, 12, 15 — increasing by 3 each time, not constant, so it is not arithmetic; the ratios are not constant either, so it is not geometric. The gaps themselves form an arithmetic sequence; the next gap is 15+3=18, so the next term is 45+18=63.

40. 18 — alternating +6 and −2: 4+6=10, 10−2=8, 8+6=14, 14−2=12, 12+6=18.

41. These totals equal n(n+1) — twice the triangular numbers (0, 2, 6, 12, 20 correspond to n=0,1,2,3,4). The 6th total (n=5) is 5×6=30.

42. A, B, D, E — each term is half the previous (A), the next term is 10÷2=5 (B), this is geometric with ratio 1/2 (D, E); it is not arithmetic, so C is false.

43. 4, 7, 12 — term(n)=term(n−1)+term(n−2)+1: term3=2+1+1=4, term4=4+2+1=7, term5=7+4+1=12.

44. 34 — this IS a genuine Fibonacci-style sequence (each term is the sum of the two before it): 13+21=34.

45. 20 — the gaps between consecutive totals are 3, 4, 5 (increasing by 1 each time); the next gap is 6, so the number of diagonals for an 8-sided polygon is 14+6=20.

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